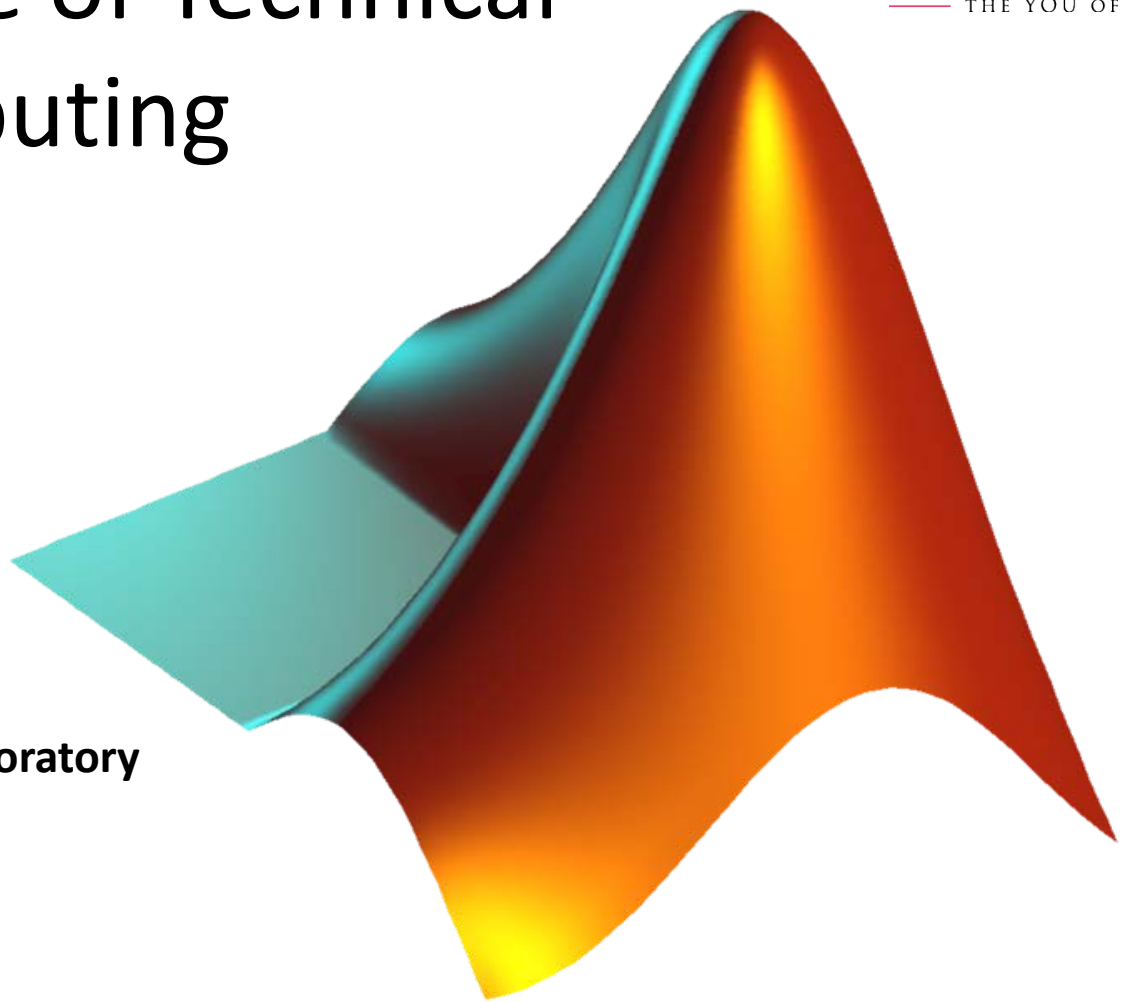
MATLAB
The Language of Technical Computing



UNIVERSITY OF ARKANSAS
THE YOU OF A

ELEG3124
System & Signal Analysis Laboratory

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What's MATLAB?

- **MATLAB** (**MAT**rix **LAB**oratory)
- A numerical & symbolic computing environment and a programming language
- Developed by MathWorks Inc.

What can MATLAB do?

- matrix manipulations
- plotting of data and functions
- implementation of algorithms
- interfacing with programs written in other languages, including C, C++, Java, and Fortran

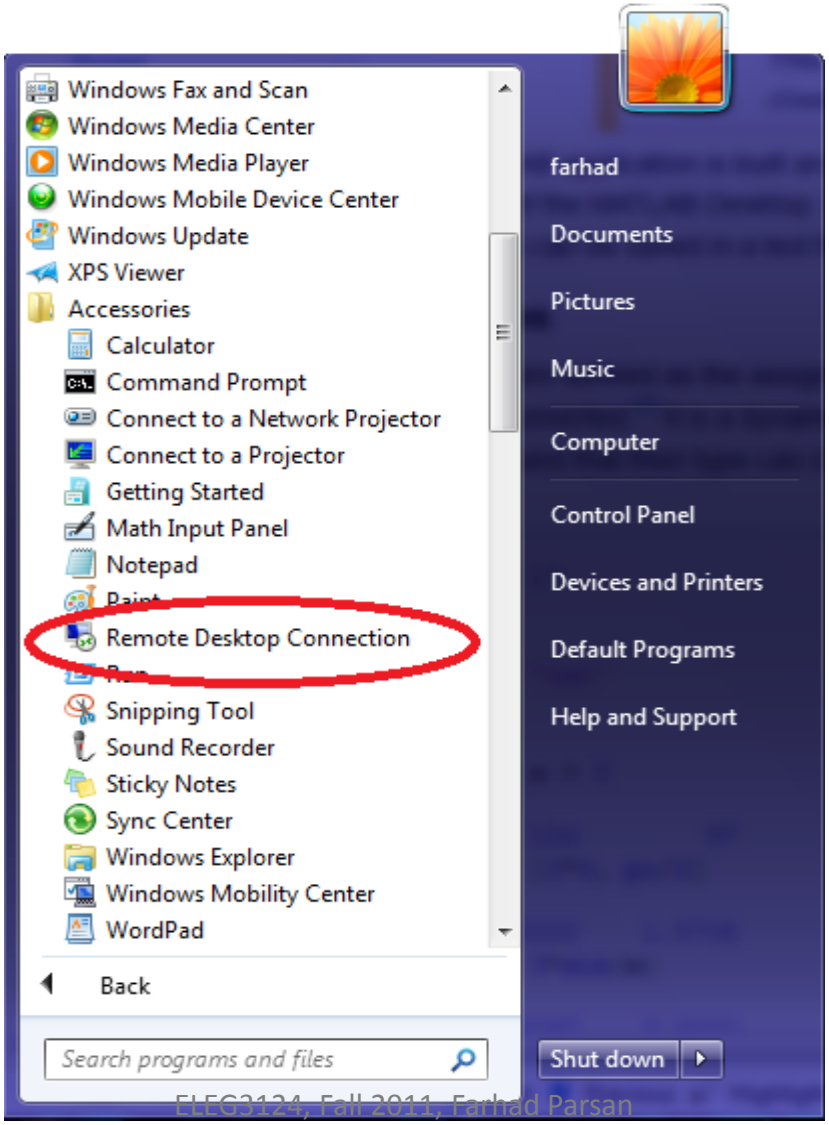
Where is MATLAB used?

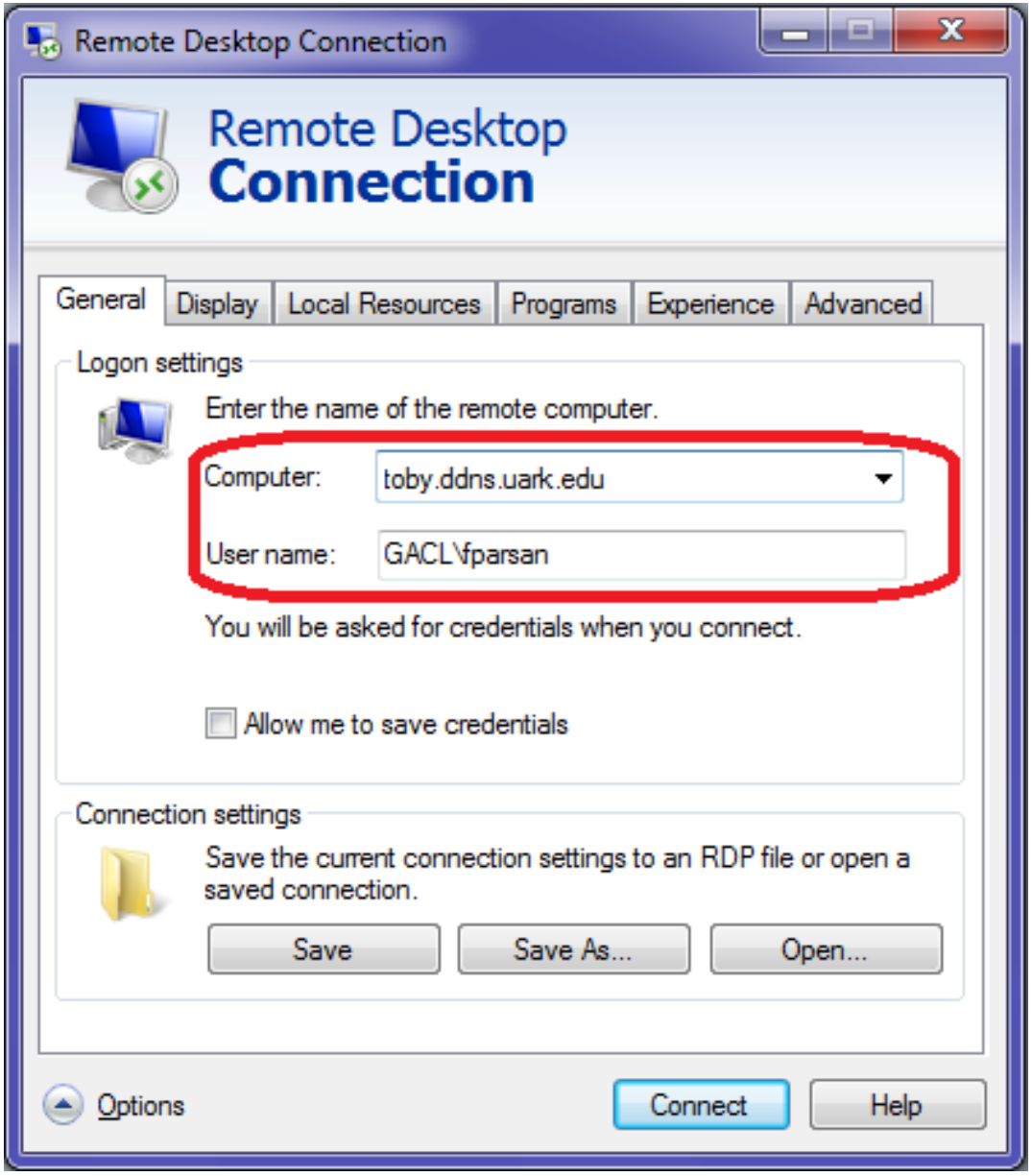
- Engineering
- Science
- Economics

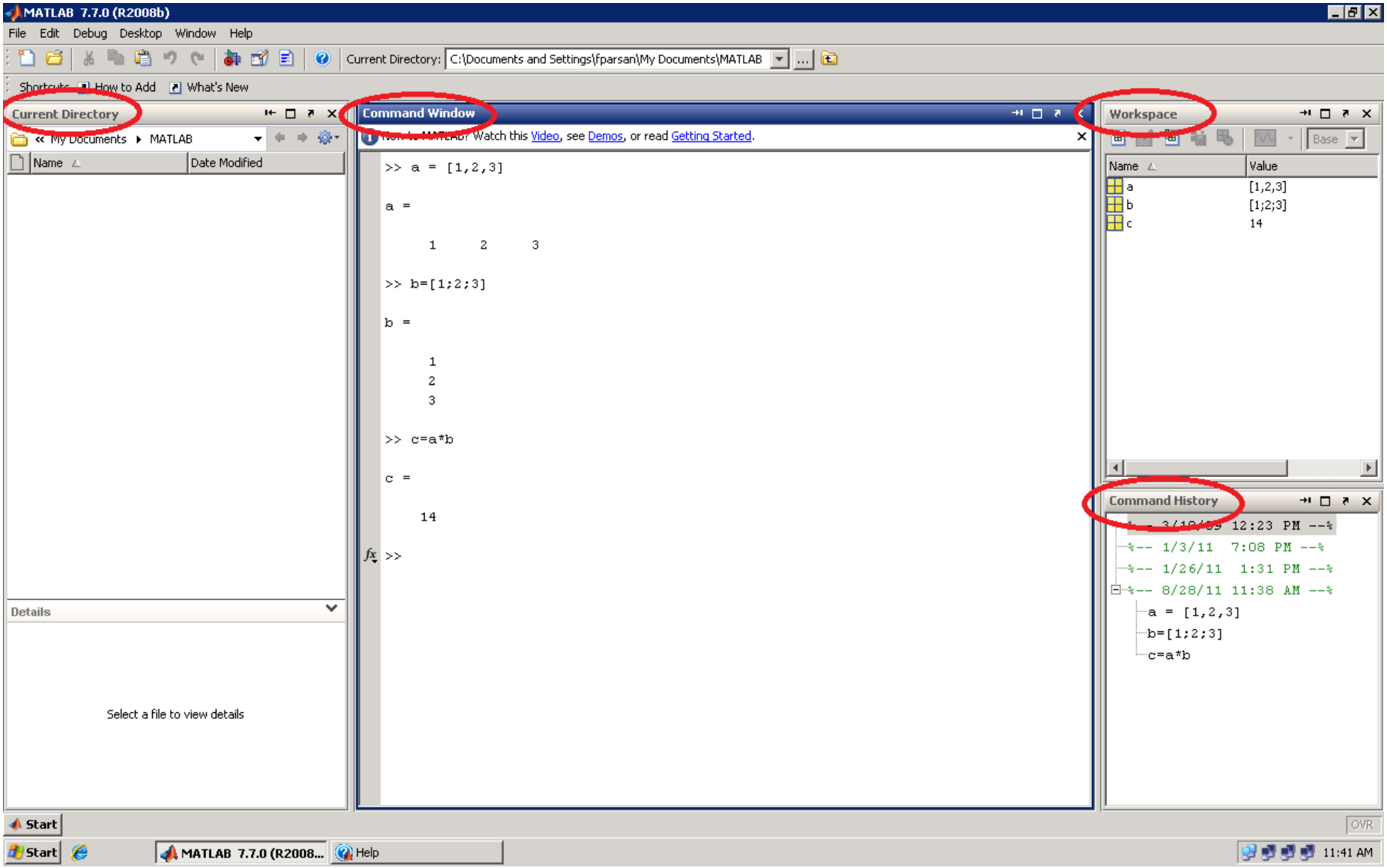
MATLAB has a very rich collection of functions used in numerous applications

MATLAB is widely used in academic and research institutions as well as industrial enterprises

Running MATLAB




Remote Desktop Connection
Remote Desktop
Connection
General
Display
Local Resources
Programs
Experience
Advanced
Logon settings
Enter the name of the remote computer.
Computer:
toby.ddns.uark.edu
User name:
GACL\fparsan
You will be asked for credentials when you connect.
Allow me to save credentials
Connection settings
Save the current connection settings to an RDP file or open a saved connection.
Save
Save As...
Open...
Options
Connect
Help


MATLAB 7.7.0 (R2008b)
File Edit Debug Desktop Window Help
Current Directory: C:\Documents and Settings\fparsan\My Documents\MATLAB
Shortcuts How to Add What's New
Current Directory
My Documents MATLAB
Name Date Modified
Details
Select a file to view details
Command Window
Watch this Video, see Demos, or read Getting Started.
>> a = [1,2,3]
a =
1 2 3
>> b=[1;2;3]
b =
1
2
3
>> c=a*b
c =
14
>>
Workspace
Base
Name Value
a [1,2,3]
b [1;2;3]
c 14
Command History
%-- 1/3/11 7:08 PM --%
%-- 1/26/11 1:31 PM --%
%-- 8/28/11 11:38 AM --%
a = [1,2,3]
b=[1;2;3]
c=a*b
Start
MATLAB 7.7.0 (R2008...
Help
11:41 AM

Vectors/Matrices

- MATLAB treats all variables as matrices
- Vectors are special forms of matrices and contain only one row OR one column
- Scalars are matrices with only one row AND one column

Vectors/Matrices Examples

- 1
- 1+2
- a = [1 2 3]
- b = [4,5,6]
- c = a+b
- d = a+b;
- d
- who
- whos
- clear d
- clear all

Vectors/Matrices

- ans is a special variable in MATLAB which always stores the result of an expression if it's not explicitly assigned to a variable
- The answer will not be displayed when a semicolon (;) is put at the end of an expression
- who shows which variables are already defined
- whos shows more details than who
- clear variable clears the specified variable from the workspace
- clear all clears all the variables from the workspace

Vectors/Matrices Algebra

- a = [1 2 3]
- b = 2*a
- c = a/0.5
- d = a-1
- Pi
- i
- j
- save myvars a b
- clear all
- whos
- load myvars a b

+ addition
- subtraction
*** multiplication**
/ division

Vectors/Matrices

- Some variable such as pi, i, j, … are predefined in MATLAB
- save filename variables saves the specified variables in a file named filename in the working directory
- load filename variables loads the selected variables from filename
- for save and load command if variables are not specified then all the variables in the workspace will be saved/loaded

Vectors/Matrices Algebra

- a = [1 2 3; 4 5 6; 7 8 9]
- b = [1,1,1;2,2,2;3,3,3]
- c = a + b
- d = a – b
- a * b
- a .* b
- a / b
- a ./ b

+ addition
- subtraction
.* multiplication
*** multiplication**
./ division
/ division

Vectors/Matrices

- Period (.) before an operation tells MATLAB to do it on individual elements than the whole matrix. For example, A*B multiplies matrix A to matrix B but A.*B multiplies each element of matrix A to the similar element of matrix B
- For addition and subtraction, both type of operations are essentially the same so there is no need to make a distinction therefore no period is allowed to be used before + or -

Vectors/Matrices Manipulation

- a(1,1)
- a(1,1)= 5
- a(1,:)
- a(:,1)
- a(1:2,1:2)
- a(1,:)= [2,2,2]
- b = 0:5
- b = 0:2:10
- c = linspace(0,10,6)
- c = logspace(1,4,4)
- clc

Vectors/Matrices

- Matrix (vector) indexes start from 1
- i:j creates a vector starting from i to j
- i:k:j creates a vector starting from i to j by increments of k
- linspace(i,j,n) creates n linearly spaced points between i and j
- logspace(i,j,n) creates n linearly spaced points between 10^i and 10^j
- clc command clears the screen

Special Matrices

- ones(i,j) creates an ixj matrix in which all the elements are 1
- zeros(i,j) creates an ixj matrix in which all the elements are 0
- rand(i,j) creates an ixj matrix in which the elements are random numbers between 0 and 1
- eye(i) creates an ixi identity matrix
- magic(i) creates an ixi magic matrix
- det(A) calculates the determinant of A
- inv(A) calculates the inverse of matrix A
- A' gives the transpose of A
- [A B] concatenates A to B
- length(A) gives the length of a vector
- size(A) gives the size of a matrix

Plotting

- plot(x,y) Plot of x versus y.
- plot(y) Plots columns of y versus their index.
- plot(x,y,'s') Plots x versus y according to rules outlined by s.
- grid on/off Adds/removes grid to current figure.
- title('text') Adds title text to current figure.
- xlabel('text') Adds x-axis label text to current figure.
- ylabel('text') Adds y-axis label text to current figure.
- hold on/off Holds current figure as is so subsequent plotting commands add to existing graph.

Plotting example

- x = -pi:.1:pi;
- y = sin(x);
- plot(x,y)
- grid on
- xlabel('X')
- ylabel('Y')
- title('sine wave')
- z = cos(x);
- plot(x,z,'r*')
- hold on
- plot(x,y,'b-.')

More Plotting Commands

- clf clears the current figure
- stem(x,y) plot of x versus y for discrete data
- figure(k) creates a new figure with number k
- close(k) closes the figure with number k
- subplot(mnk) divides the figure area to mxn sections and plots the subsequent commands in section k
- axis([XMIN XMAX YMIN YMAX]) assign the axis limit manually
- xlim([XMIN XMAX]) assign the x-axis limit manually
- ylim([YMIN YMAX]) assign the y-axis limit manually
- legend('text') creates a legend for the current plot

Symbolic Math

- syms alerts MATLAB that you are using a symbolic variable, and that it does not have a specific value.
- ezplot plots symbolic functions
- subs substitutes a number or a symbol into a symbolic expression
- simplify simplifies a symbolic expression
- int calculates the symbolic integration
- diff calculates the symbolic differentiation

Symbolic Math Examples

- syms x y
- y = cos(x)
- ezplot(y,[0:0.1:2*pi])
- a = [0:0.1:2*pi];
- b = subs(y,x,a);
- plot(a,b)
- y = (x^2+5*x+6)/(x+2)
- z = simplify (y)
- diff(z)
- int(z)

Programming

- Expressions
- Flow Control blocks
  - Conditional
  - Iterations
- Scripts
- Functions

Programming Operators

Operator	Description
<	Less than
<=	Less than or equal to
>	Greater than
>=	Greater than or equal to
==	Equal to
~=	Not equal to

Logical Operation	Description
A && B	Logical and
A || B	Logical or

Programming Flow Control

```
if expression1
   statements1
elseif
expression2
   statements2
else
   statements3
end
```

```
for variable = initval:endval
   statement
   ...
   statement
end
```

```
while expression
   statements
end
```

Programming Examples

```
a = input('input a? ');
b = input('input b? ');
if a == b,
   fprintf('a is equal to b\n');
elseif a > 0 && b > 0
   fprintf('both positive\n');
else
   fprintf('other case\n');
end
```

```
M = rand(10,10); suma = 0;
for i = {2,5:8}
   for j = {1:5,8:9}
      suma = suma + M(i,j);
   end
end
fprintf('sum = %d\n',suma);
```

```
M = rand(4,4);
i = 1; j = 1; suma = 0;
while i <= 4
   while j <= 4
      suma = suma + M(i,j);
      j = j+1;
   end
i = i+1;
end
fprintf('suma = %f\n',suma);
```

```
M = rand(4,4); suma = 0;
for i = 1:4
   for j = 1:4
      suma = suma + M(i,j);
   end
end
fprintf('sum = %d\n',suma);
```

Creating an m-file

- scripts/functions are usually referred to as m-file in MATLAB because they are saved with *.m extension
- The name of an M-file must begin with an alphabetic character
- scripts are simply pieces of code that are organized inside a file (i.e. they could be run line-by-line inside the MATLAB workspace) so they can add to workspace variables
- functions cannot be run separately but they have to be called inside other scripts/functions and their variables are local

Writing your first m-file (script)

```
% This script creates a NxN multiplication table
clc
clear all
N = input('Please enter size : ');
if (N > 10)
   disp('N must be less than 10');
else
   for i=1:N
      for j=1:N
         z = i*j;
         fprintf('%5d',z);
      end
      fprintf('\n');
   end
end
```

Writing your first m-file (function)

```
% This function adds the divisors of N
% e.g. N = 10 -> 1+2+5 = 8
function result = divisor_sum(N)
sum = 0;
for i=1:N-1
   if mod(N,i)== 0
      sum=sum+i;
   end
end
result = sum;
```

References

- http://en.wikipedia.org/wiki/MATLAB
- http://www.mathworks.com/index.html
- http://www.karenkopecky.net/Teaching/eco613614/Matlab%20Resources/MatlabCheatSheet.pdf
- http://www.cs.jhu.edu/CIRL/class/600.161/class2/shortmatlab.pdf